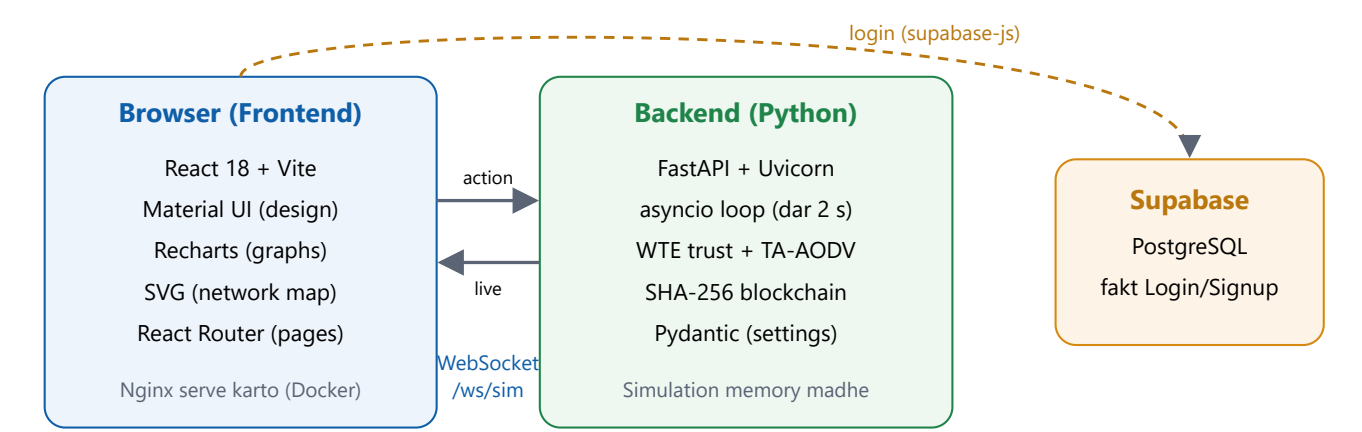


TrustChain-WSN — Technologies Used

Project madhe kontya technologies vaparlya, kuthe ani ka · Sagli mahiti `backend/requirements.txt`, `frontend/package.json` ani code madhun

Ekach olit: Backend **Python FastAPI + WebSocket**, frontend **React + Vite + Material UI + Recharts**, blockchain **SHA-256 Proof-of-Work** (swatah lihilela), login sathi **Supabase**, ani deployment sathi **Docker + Nginx**.

1. Architecture — kon kay karto



Docker Compose: backend port 8001 · frontend port 5174

Backend ha project cha **"brain"** aahe — sagli simulation, trust calculation, routing ani blockchain tithech chalte. Frontend fakt **display** aahe — backend je pathavto te dakhavto. Supabase cha simulation shi kahihi sambandh nahi; to fakt login sathi.

2. Backend Technologies

Technology	Kuthe vaparli	Ka nivadli
Python 3.11	Sagla backend code	Simulation ani algorithms sathi sopi, vachayla saral language; research madhe khup vaparli jate
FastAPI	REST APIs: <code>/api/nodes</code> , <code>/api/attacks</code> , <code>/api/ledger</code> , <code>/api/routes</code> , <code>/api/metrics</code>	Fast, modern, automatic API documentation (<code>/docs</code>), WebSocket support built-in
Uvicorn	FastAPI app run karnara server (port 8001)	Async (ASGI) server — WebSocket ani background loop ekach veli chalu shakto
WebSockets	<code>/ws/sim</code> — dar round nantar live data browser la push	Browser la vicharayla nako (polling nahi) — server swatah update pathavto, mhanun dashboard real-time

asyncio	Background simulation loop — dar 2 second la ek round	Browser ughadla nasla tari simulation chalu rahte, industrial monitoring system sarkhi
Pydantic + pydantic-settings	Data schemas ani <code>config.py</code> (threshold 0.40, quarantine, battery costs)	Sagle settings ekach thikani; <code>.env</code> file ne badalta yetat
python-dotenv	<code>.env</code> file madhun keys/settings load	Secret keys code madhe lihayla nakot
hashlib (SHA-256)	<code>blockchain.py</code> — block hash, Proof-of-Work mining, Merkle root	Python madhe built-in; blockchain swatah lihilela aahe, Ethereum/Hyperledger sarkhi bahya library nahi — lightweight theva yeto
supabase (Python)	<code>supabase_client.py</code>	Database connection (login sathi)

3. Frontend Technologies

Technology	Kuthe vaparli	Ka nivadli
React 18	Sagli 12 pages: Dashboard, Network Map, Trust Engine, Attack Lab, Blockchain Ledger, Secure Routing, Performance, Settings...	Component-based — ekda banavlela panel anek pages var vaparta yeto; live data aala ki screen automatically update
Vite 5	Dev server (port 5173) ani production build	Khup fast start ani build
Material UI (MUI 5) + Emotion	Cards, tables, buttons, chips, icons, dark/light theme	Professional design ready-made; dark mode sopa
Recharts	Sagle graphs: PDR, throughput, delay, energy vs overhead, recovery time	React sathi banavleli chart library — live data var smooth update
SVG	Network Map — nodes, routes, wormhole tunnel, quarantine rings, battery bars	Zoom kela tari clear; pratyek node la click/hover karta yeto
React Router 6	Pages madhe navigation (<code>/dashboard</code> , <code>/topology</code> ...), login protection	Single Page App — page reload nahi
Axios + TanStack React Query	REST API calls (<code>services/api.js</code>)	API data fetch ani cache sathi
supabase-js	Login / Register pages	Ready-made authentication
Inter, JetBrains Mono fonts	Normal text ani node IDs / hashes	Vachayla clear; hash ani IDs monospace madhe neat distat

4. Database ani Login — Supabase

- **Supabase** = online PostgreSQL database + authentication service.
- Project madhe fakt **user login / signup** sathi vaparla.
- Simulation cha data (nodes, attacks, blocks) database madhe **save hot nahi** — to backend chya memory madhe asto. Backend restart kela ki navin random network tayar hoto.
- SQL files: `supabase/schema.sql` , `seed.sql` .

5. Deployment Technologies

Technology	Kuthe vaparli	Ka
Docker	<code>backend/Dockerfile</code> , <code>frontend/Dockerfile</code>	Konatyahi computer var sarkhech chalte — "majhya PC var chalte" problem nahi
Docker Compose	<code>docker-compose.yml</code> — ekach command ne donhi chalu (<code>docker compose up --build</code>)	Backend (8001) ani frontend (5174) ekatra
Nginx	<code>frontend/nginx.conf</code> — built frontend serve karto	Fast, production-grade web server
Vercel	<code>frontend/vercel.json</code>	Frontend online host karaycha option
START.bat	Windows var ek click ne project chalu	Demo sathi sopa

6. Algorithms (swatah implement kele — library nahi)

Algorithm	Kaam	File
Weighted Trust Evaluation (WTE)	Attack detection — drop, delay, identity var trust penalty	<code>sim/network.py</code>
Trust-Aware AODV (TA-AODV)	Secure routing — distance ÷ trust, bad nodes talne	<code>sim/network.py</code>
Second-Chance Redemption	Automatic recovery — quarantine → scrub → probation → readmit	<code>sim/network.py</code>
Event-Triggered PoW	Blockchain — fakt attack var block, SHA-256, Merkle tree, difficulty 2	<code>sim/blockchain.py</code>

7. Development Tools

- **VS Code** — code editor
- **Git** — version control
- **npm** — frontend packages; **pip** + **venv** — Python packages
- **ESLint** — JavaScript code check

8. Viva madhe vicharle jaau shaktat ase prashna

Prashna	Uttar
Blockchain sathi konti library?	Konatihi nahi — Python <code>hashlib</code> (SHA-256) vaparun swatah lihilela lightweight blockchain. Ethereum sarkha heavy blockchain sensor network sathi yogya nahi.
WebSocket ka, REST ka nahi?	REST madhe browser la vicharat rahava lagte (polling). WebSocket madhe server swatah dar round la data pathavto — real-time, kami traffic.
FastAPI ka, Flask/Django ka nahi?	FastAPI async aahe — WebSocket ani background simulation loop ekach veli chalavta yetat; automatic API docs.
React ka?	Live data badalla ki fakt to bhag update hoto; components parat vaparta yetat.
Database konta?	Supabase (PostgreSQL) — fakt login sathi. Simulation data memory madhe.
Graphs kashane?	Recharts library; network map SVG ne.
Project kasa run karaycha?	Backend: <code>uvicorn app.main:app --port 8001</code> · Frontend: <code>npm run dev</code> · Kiva <code>docker compose up --build</code>
Real sensors vaparle ka?	Nahi — he simulation aahe. Nodes, packets, battery he Python madhe model kele aahet; real hardware nahi.

Lakshat theva: `requirements.txt` madhe **numpy**, **python-jose**, **httpx** pan aahet, pan code madhe te vaparle nahit. Viva madhe yanna "vaparleli technology" mhanun sangu naka.